

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup $>1\times$ means SoA is faster; **bold** marks $>1.1\times$.

Program	ADT fields	SoA bufs	Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
Compiler	9	8	0.1403	0.0668	2.10 $\times$	0.1622	0.1806	0.90 $\times$
DBQuery	15	13	0.1375	0.2199	0.63 $\times$	0.1371	0.1692	0.81 $\times$
DecisionTree	7	6	3.984	3.335	1.19 $\times$	—	—	—
DomTree	15	14	0.1396	0.0616	2.26 $\times$	0.0585	0.0805	0.73 $\times$
KDTree	17	16	0.1823	0.1705	1.07 $\times$	—	—	—
LinearListReduction	11	11	0.0301	0.0063	4.77 $\times$	—	—	—
List	2	2	0.1319	0.1111	1.19 $\times$	0.2772	0.3409	0.81 $\times$
MonoTree	3	2	0.0180	0.0225	0.80 $\times$	0.0952	0.1025	0.93 $\times$
ObjectGraph	5	4	0.0952	0.0943	1.01 $\times$	0.1453	0.1639	0.89 $\times$
OctTree_sumMass	16	—	0.0362	0.0284	1.27 $\times$	—	—	—
OctTree_sumEnergy	16	—	0.0363	0.0336	1.08 $\times$	—	—	—
OctTree_countActive	16	—	0.0351	0.0332	1.06 $\times$	—	—	—
OctTree_countParticles	16	—	0.0355	0.0306	1.16 $\times$	—	—	—
OctTree_barnesHutPotential	16	—	0.0417	0.0451	0.92 $\times$	—	—	—
OctTree_fmmPotential	16	—	0.0974	0.1257	0.77 $\times$	—	—	—
OctTree_scaleEnergy	16	—	—	—	—	0.1523	0.1768	0.86 $\times$
OctTree_clearFlags	16	—	—	—	—	0.1535	0.1975	0.78 $\times$
PiecewiseFunctions	8	7	0.1382	0.2321	0.60 $\times$	0.2391	0.2670	0.90 $\times$
TernaryTree	5	3	0.0290	0.0277	1.05 $\times$	0.0836	0.0882	0.95 $\times$
Trie	10	9	0.0622	0.0481	1.29 $\times$	0.1498	0.1694	0.88 $\times$
ColorOctree	25	18	0.0944	0.1029	0.92 $\times$	—	—	—

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. $>1\times$ means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
Compiler	1.59 $\times$	0.52 $\times$	2.40 $\times$	0.90 $\times$	9.28 $\times$	0.53 $\times$	7.44 $\times$
DBQuery	1.08 $\times$	0.47 $\times$	1.58 $\times$	0.47 $\times$	9.00 $\times$	0.26 $\times$	5.99 $\times$
DecisionTree	1.30 $\times$	0.59 $\times$	16.69 $\times$	1.69 $\times$	1.01 $\times$	1.84 $\times$	36.01 $\times$
DomTree	1.54 $\times$	0.57 $\times$	2.10 $\times$	1.18 $\times$	19.97 $\times$	0.24 $\times$	10.13 $\times$
KDTree	1.31 $\times$	0.77 $\times$	1.14 $\times$	0.85 $\times$	0.22 $\times$	0.78 $\times$	4.98 $\times$
LinearListReduction	5.82 $\times$	0.90 $\times$	1.22 $\times$	2.04 $\times$	4.22 $\times$	3.27 $\times$	10.89 $\times$
List	0.81 $\times$	0.80 $\times$	3.37 $\times$	0.55 $\times$	0.13 $\times$	0.43 $\times$	1.46 $\times$
MonoTree	0.98 $\times$	0.85 $\times$	0.30 $\times$	0.52 $\times$	0.23 $\times$	0.84 $\times$	1.20 $\times$
ObjectGraph	0.98 $\times$	0.75 $\times$	4.69 $\times$	0.70 $\times$	0.46 $\times$	0.74 $\times$	2.81 $\times$
OctTree_sumMass	1.20 $\times$	0.89 $\times$	8.00 $\times$	1.12 $\times$	2.16 $\times$	1.13 $\times$	4.79 $\times$
OctTree_sumEnergy	1.10 $\times$	0.87 $\times$	0.29 $\times$	1.08 $\times$	0.07 $\times$	1.31 $\times$	2.35 $\times$
OctTree_countActive	1.11 $\times$	1.01 $\times$	1.76 $\times$	1.16 $\times$	0.25 $\times$	1.64 $\times$	17.19 $\times$
OctTree_countParticles	1.19 $\times$	1.00 $\times$	4.21 $\times$	1.63 $\times$	0.46 $\times$	2.21 $\times$	42.52 $\times$
OctTree_barnesHutPotential	0.95 $\times$	0.82 $\times$	0.01 $\times$	0.95 $\times$	0.01 $\times$	0.80 $\times$	3.70 $\times$
OctTree_fmmPotential	1.01 $\times$	0.87 $\times$	0.01 $\times$	0.41 $\times$	0.03 $\times$	0.35 $\times$	2.06 $\times$
OctTree_scaleEnergy	0.84 $\times$	0.49 $\times$	0.40 $\times$	0.39 $\times$	0.19 $\times$	0.58 $\times$	2.18 $\times$
OctTree_clearFlags	0.83 $\times$	0.46 $\times$	0.39 $\times$	0.36 $\times$	0.25 $\times$	1.40 $\times$	2.32 $\times$
PiecewiseFunctions	1.09 $\times$	0.60 $\times$	2.52 $\times$	0.90 $\times$	2.09 $\times$	0.41 $\times$	4.93 $\times$
TernaryTree	1.14 $\times$	0.79 $\times$	0.25 $\times$	0.50 $\times$	0.24 $\times$	1.06 $\times$	1.43 $\times$
Trie	1.11 $\times$	0.56 $\times$	1.30 $\times$	0.82 $\times$	1.37 $\times$	0.17 $\times$	6.19 $\times$
ColorOctree	1.18 $\times$	0.84 $\times$	1.86 $\times$	0.77 $\times$	0.20 $\times$	0.75 $\times$	4.60 $\times$
Geomean	1.19 $\times$	0.71 $\times$	0.96 $\times$	0.81 $\times$	0.52 $\times$	0.76 $\times$	4.87 $\times$

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Compiler	9.6e+08/ 6.0e+08	9.9e+08 /1.9e+09	1.3e+07/ 5.4e+06	3.0e+05 /3.3e+05	1.2e+06/ 1.3e+05	1.7e+04 /3.2e+04	5.1e+07/ 6.8e+06
DBQuery	9.6e+08/ 8.9e+08	1.1e+09 /2.2e+09	5.8e+06/ 3.7e+06	1.7e+05 /3.5e+05	9.3e+05/ 1.0e+05	8.6e+03 /3.2e+04	2.0e+07/ 3.4e+06
DecisionTree	2.0e+10/ 1.6e+10	4.0e+10 /6.8e+10	5.3e+08/ 3.2e+07	9.2e+04/ 5.4e+04	2.2e+06/ 2.1e+06	9.1e+04/ 5.0e+04	1.1e+09/ 3.0e+07
DomTree	8.3e+08/ 5.4e+08	1.2e+09 /2.2e+09	1.2e+07/ 5.7e+06	1.4e+05/ 1.2e+05	1.1e+06/ 5.5e+04	7.8e+03 /3.2e+04	5.7e+07/ 5.6e+06
KDTree	9.2e+08/ 7.0e+08	3.1e+09 /4.0e+09	1.4e+07/ 1.2e+07	1.6e+03 /1.9e+03	2.5e+04 /1.1e+05	1.4e+03 /1.8e+03	4.4e+07/ 8.9e+06
LinearListReduction	1.5e+08/ 2.6e+07	9.0e+07 /1.0e+08	1.5e+06/ 1.2e+06	7.8e+02/ 3.8e+02	2.8e+05/ 6.7e+04	6.7e+02/ 2.0e+02	1.2e+07/ 1.1e+06
List	1.3e+09 /1.5e+09	4.5e+09 /5.6e+09	9.5e+06/ 2.8e+06	3.4e+05 /6.1e+05	2.5e+04 /2.0e+05	5.6e+03 /1.3e+04	4.8e+07/ 3.3e+07
MonoTree	2.6e+08 /2.6e+08	9.1e+08 /1.1e+09	2.6e+05 /8.8e+05	3.5e+04 /6.7e+04	4.5e+03 /1.9e+04	2.4e+03 /2.9e+03	2.2e+06/ 1.8e+06
ObjectGraph	8.6e+08 /8.8e+08	3.0e+09 /4.0e+09	9.6e+06/ 2.1e+06	1.7e+05 /2.4e+05	3.1e+04 /6.7e+04	1.2e+04 /1.7e+04	2.4e+07/ 8.5e+06
OctTree_sumMass	1.7e+08/ 1.4e+08	5.7e+08 /6.3e+08	1.2e+06/ 1.5e+05	5.0e+02/ 4.5e+02	1.8e+04/ 8.3e+03	3.4e+02/ 3.0e+02	6.2e+06/ 1.3e+06
OctTree_sumEnergy	1.8e+08/ 1.7e+08	7.7e+08 /8.9e+08	3.4e+05 /1.2e+06	5.5e+02/ 5.1e+02	2.4e+03 /3.4e+04	4.6e+02/ 3.5e+02	6.3e+06/ 2.7e+06
OctTree_countActive	1.8e+08/ 1.6e+08	5.8e+08/ 5.8e+08	8.4e+05/ 4.8e+05	5.4e+02/ 4.7e+02	3.9e+03 /1.5e+04	4.8e+02 / 3.0e+02	6.3e+06/ 3.7e+05
OctTree_countParticles	1.8e+08/ 1.5e+08	5.7e+08/ 5.7e+08	8.6e+05/ 2.0e+05	6.0e+02/ 3.6e+02	3.9e+03 /8.4e+03	4.6e+02/ 2.1e+02	6.2e+06/ 1.4e+05
OctTree_barnesHutPotential	2.1e+08 /2.2e+08	8.2e+08 /1.0e+09	7.1e+04 /5.7e+06	5.1e+02 /5.4e+02	1.6e+03 /1.1e+05	3.3e+02 /4.1e+02	6.3e+06/ 1.7e+06
OctTree_fmmPotential	4.9e+08/ 4.9e+08	1.6e+09 /1.8e+09	2.5e+04 /2.5e+06	5.1e+02 /1.2e+03	1.8e+03 /6.0e+04	3.6e+02 /1.0e+03	4.5e+06/ 2.2e+06
OctTree_scaleEnergy	3.6e+08 /4.3e+08	9.1e+08 /1.9e+09	1.2e+06 /2.9e+06	1.6e+05 /4.1e+05	8.9e+03 /4.6e+04	6.7e+03 /1.1e+04	5.7e+06/ 2.6e+06
OctTree_clearFlags	3.7e+08 /4.4e+08	8.0e+08 /1.7e+09	9.5e+05 /2.5e+06	2.1e+05 /5.7e+05	9.7e+03 /3.9e+04	5.2e+03/ 3.7e+03	5.6e+06/ 2.4e+06
PiecewiseFunctions	1.3e+09/ 1.2e+09	2.8e+09 /4.7e+09	1.4e+07/ 5.6e+06	4.2e+05 /4.7e+05	1.9e+05/ 9.2e+04	2.0e+04 /4.9e+04	4.5e+07/ 9.2e+06
TernaryTree	4.2e+08/ 3.7e+08	1.2e+09 /1.5e+09	3.9e+05 /1.5e+06	6.6e+04 /1.3e+05	5.1e+03 /2.1e+04	2.8e+03/ 2.7e+03	3.4e+06/ 2.3e+06
Trie	6.4e+08/ 5.8e+08	1.2e+09 /2.2e+09	8.5e+06/ 6.5e+06	1.8e+05 /2.2e+05	1.6e+05/ 1.2e+05	6.9e+03 /4.1e+04	2.6e+07/ 4.3e+06
ColorOctree	4.8e+08/ 4.0e+08	1.6e+09 /2.0e+09	9.4e+06/ 5.1e+06	7.1e+02 /9.2e+02	2.6e+04 /1.3e+05	5.8e+02 /7.7e+02	1.9e+07/ 4.0e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.3025	1.122	0.2474	<i>OOM</i>
DBQuery	0.2746	0.3385	0.3891	1.248
DecisionTree	3.984	5.879	3.335	24.96
DomTree	0.1980	0.3620	0.1421	0.5538
KDTree	0.1823	0.2692	0.1705	0.7294
LinearListReduction	0.0301	0.1340	0.0063	<i>OOM</i>
List	0.4091	<i>OOM</i>	0.4521	<i>OOM</i>
MonoTree	0.1132	0.2964	0.1249	0.3224
ObjectGraph	0.2405	0.5523	0.2581	0.7104
OctTree_sumMass	0.0362	0.0461	0.0284	<i>OOM</i>
OctTree_sumEnergy	0.0363	0.0562	0.0336	<i>OOM</i>
OctTree_countActive	0.0351	0.0757	0.0332	<i>OOM</i>
OctTree_countParticles	0.0355	0.0461	0.0306	<i>OOM</i>
OctTree_barnesHutPotential	0.0417	0.0612	0.0451	<i>OOM</i>
OctTree_fmmPotential	0.0974	0.2106	0.1257	<i>OOM</i>
OctTree_scaleEnergy	0.1523	0.1679	0.1768	<i>OOM</i>
OctTree_clearFlags	0.1535	0.1537	0.1975	<i>OOM</i>
PiecewiseFunctions	0.3773	0.5654	0.4990	2.201
TernaryTree	0.1126	0.2073	0.1159	0.6487
Trie	0.2120	0.3114	0.2175	0.9972
ColorOctree	0.0944	0.1182	0.1029	<i>OOM</i>

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.22 $\times$	3.71 $\times$	4.53 $\times$	–
DBQuery	0.71 $\times$	1.23 $\times$	0.87 $\times$	0.27 $\times$
DecisionTree	1.19 $\times$	1.48 $\times$	1.76 $\times$	0.24 $\times$
DomTree	1.39 $\times$	1.83 $\times$	2.55 $\times$	0.65 $\times$
KDTree	1.07 $\times$	1.48 $\times$	1.58 $\times$	0.37 $\times$
LinearListReduction	4.77 $\times$	4.45 $\times$	21.23 $\times$	–
List	0.90 $\times$	–	–	–
MonoTree	0.91 $\times$	2.62 $\times$	2.37 $\times$	0.92 $\times$
ObjectGraph	0.93 $\times$	2.30 $\times$	2.14 $\times$	0.78 $\times$
OctTree_sumMass	1.27 $\times$	1.27 $\times$	1.62 $\times$	–
OctTree_sumEnergy	1.08 $\times$	1.55 $\times$	1.67 $\times$	–
OctTree_countActive	1.06 $\times$	2.15 $\times$	2.28 $\times$	–
OctTree_countParticles	1.16 $\times$	1.30 $\times$	1.50 $\times$	–
OctTree_barnesHutPotential	0.92 $\times$	1.47 $\times$	1.36 $\times$	–
OctTree_fmmPotential	0.77 $\times$	2.16 $\times$	1.68 $\times$	–
OctTree_scaleEnergy	0.86 $\times$	1.10 $\times$	0.95 $\times$	–
OctTree_clearFlags	0.78 $\times$	1.00 $\times$	0.78 $\times$	–
PiecewiseFunctions	0.76 $\times$	1.50 $\times$	1.13 $\times$	0.26 $\times$
TernaryTree	0.97 $\times$	1.84 $\times$	1.79 $\times$	0.32 $\times$
Trie	0.97 $\times$	1.47 $\times$	1.43 $\times$	0.31 $\times$
ColorOctree	0.92 $\times$	1.25 $\times$	1.15 $\times$	–

Table 6: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0122	0.0800	0.0037	<i>OOM</i>	3.34 $\times$	6.56 $\times$	21.91 $\times$	–
branchStatsPass	F	2/9	78%	0.0120	0.0804	0.0046	<i>OOM</i>	2.62 $\times$	6.72 $\times$	17.59 $\times$	–
castInstCountPass	F	3/9	67%	0.0185	0.0806	0.0020	<i>OOM</i>	9.17 $\times$	4.35 $\times$	39.84 $\times$	–
has cycle	F	4/9	56%	0.0353	0.0943	0.0325	<i>OOM</i>	1.08 $\times$	2.68 $\times$	2.90 $\times$	–
instCountPass	F	2/9	78%	0.0126	0.0797	0.0040	<i>OOM</i>	3.13 $\times$	6.32 $\times$	19.75 $\times$	–
latencyModelPass	F	3/9	67%	0.0122	0.0842	0.0034	<i>OOM</i>	3.53 $\times$	6.91 $\times$	24.41 $\times$	–
memoryOpStatsPass	F	3/9	67%	0.0121	0.0803	0.0065	<i>OOM</i>	1.86 $\times$	6.65 $\times$	12.34 $\times$	–
stripSideEffectsPass	M	7/9	22%	0.0802	0.1830	0.0890	<i>OOM</i>	0.90 $\times$	2.28 $\times$	2.06 $\times$	–
targetReturnPass	M	9/9	0%	0.0820	0.2005	0.0916	<i>OOM</i>	0.89 $\times$	2.45 $\times$	2.19 $\times$	–
throughputModelPass	F	3/9	67%	0.0122	0.0806	0.0030	<i>OOM</i>	4.06 $\times$	6.61 $\times$	26.83 $\times$	–
verifyIR	F	9/9	0%	0.0133	0.0784	0.0071	<i>OOM</i>	1.89 $\times$	5.88 $\times$	11.11 $\times$	–
Total				0.3025	1.122	0.2474	–	1.22 $\times$	3.71 $\times$	4.53 $\times$	–
Geomean								2.33 $\times$	4.82 $\times$	11.22 $\times$	–

Table 7: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	6.2e+07/ 9.9e+06	5.1e+07/ 4.5e+07	1.2e+06/ 1.0e+05	3.1e+02/ 8.2e+01	2.2e+05/ 2.6e+03	2.8e+02/ 6.4e+01	4.7e+06/ 1.3e+04
branchStatsPass	F	6.0e+07/ 1.6e+07	7.9e+07 /8.6e+07	8.1e+05/ 2.2e+04	4.8e+02/ 1.0e+02	5.2e+04/ 3.3e+03	3.8e+02/ 1.0e+02	4.7e+06/ 4.9e+05
castInstCountPass	F	9.4e+07/ 6.6e+06	5.1e+07/ 3.9e+07	1.4e+04 /3.2e+04	2.6e+02/ 8.9e+01	9.0e+02 /1.1e+03	2.0e+02/ 7.5e+01	4.7e+06/ 1.8e+04
has cycle	F	1.3e+08/ 9.0e+07	1.1e+08 /1.5e+08	2.2e+06/ 1.2e+06	8.2e+04/ 5.8e+04	1.1e+05/ 2.2e+04	2.2e+03/ 6.6e+02	5.0e+06/ 1.5e+06
instCountPass	F	6.4e+07/ 1.0e+07	5.6e+07 /5.7e+07	1.2e+06/ 1.0e+05	7.6e+02/ 1.1e+02	2.2e+05/ 2.7e+03	5.8e+02/ 9.7e+01	4.7e+06/ 3.5e+04
latencyModelPass	F	6.2e+07/ 1.3e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.5e+04	2.8e+02/ 8.7e+01	2.1e+05/ 4.0e+03	2.6e+02/ 6.4e+01	4.8e+06/ 6.3e+05
memoryOpStatsPass	F	6.1e+07/ 1.9e+07	8.5e+07 /9.1e+07	6.6e+05/ 5.6e+05	6.7e+02/ 1.1e+02	3.1e+04/ 2.0e+04	4.4e+02/ 9.5e+01	4.7e+06/ 4.2e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	1.9e+06/ 1.1e+06	1.0e+05 /1.4e+05	7.3e+03 /2.0e+04	3.4e+03 /7.5e+03	4.2e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /2.1e+08	1.7e+08 /6.3e+08	1.6e+06/ 1.4e+06	1.1e+05 /1.3e+05	1.1e+04 /2.4e+04	7.1e+03 /2.2e+04	3.7e+06/ 1.4e+06
throughputModelPass	F	6.2e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.4e+04	8.4e+02/ 1.2e+02	2.1e+05/ 3.6e+03	7.6e+02/ 8.2e+01	4.8e+06/ 6.9e+05
verifyIR	F	6.0e+07/ 1.7e+07	7.4e+07 /8.6e+07	9.1e+05/ 7.8e+05	1.0e+03/ 7.3e+02	1.4e+05/ 2.8e+04	9.4e+02/ 5.4e+02	4.7e+06/ 4.2e+05
Total		9.6e+08/6.0e+08	9.9e+08/1.9e+09	1.3e+07/5.4e+06	3.0e+05/3.3e+05	1.2e+06/1.3e+05	1.7e+04/3.2e+04	5.1e+07/6.8e+06

Table 8: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.1e+08/−	1.6e+08/−	2.3e+06/−	5.7e+02/−	9.1e+04/−	5.5e+02/−	1.4e+07/−
branchStatsPass	F	4.1e+08/−	1.8e+08/−	2.1e+06/−	5.2e+02/−	4.4e+04/−	4.9e+02/−	1.4e+07/−
castInstCountPass	F	4.1e+08/−	1.8e+08/−	2.1e+06/−	6.1e+02/−	4.7e+04/−	5.7e+02/−	1.4e+07/−
has cycle	F	4.8e+08/−	2.1e+08/−	2.5e+06/−	3.8e+02/−	7.0e+04/−	3.6e+02/−	1.7e+07/−
instCountPass	F	4.0e+08/−	1.6e+08/−	2.3e+06/−	4.4e+02/−	9.8e+04/−	4.2e+02/−	1.4e+07/−
latencyModelPass	F	4.3e+08/−	1.7e+08/−	2.1e+06/−	3.8e+02/−	4.6e+04/−	3.6e+02/−	1.4e+07/−
memoryOpStatsPass	F	4.1e+08/−	1.9e+08/−	2.1e+06/−	3.7e+02/−	4.3e+04/−	3.4e+02/−	1.4e+07/−
stripSideEffectsPass	M	6.7e+08/−	3.3e+08/−	8.0e+05/−	1.0e+05/−	1.6e+04/−	1.1e+04/−	2.4e+07/−
targetReturnPass	M	6.5e+08/−	3.2e+08/−	1.1e+06/−	1.2e+05/−	1.9e+04/−	1.9e+04/−	2.2e+07/−
throughputModelPass	F	4.1e+08/−	1.7e+08/−	2.1e+06/−	3.7e+02/−	4.5e+04/−	3.5e+02/−	1.4e+07/−
verifyIR	F	3.1e+08/−	1.2e+08/−	2.2e+06/−	5.7e+02/−	1.8e+05/−	4.8e+02/−	1.1e+07/−
Total		5.0e+09/−	2.2e+09/−	2.2e+07/−	2.3e+05/−	7.0e+05/−	3.4e+04/−	1.7e+08/−

Table 9: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); ± shows standard error of the mean across −iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0680	0.0749	0.0836	0.4322	0.81×	1.10 ×	0.90×	0.17×
countJoins	F	3/15	80%	0.0200	0.0291	0.0120	0.0849	1.66 ×	1.46 ×	2.42 ×	0.34×
filterSelectivitySkew	F	5/15	67%	0.0200	0.0283	0.0280	0.0861	0.72×	1.41 ×	1.01×	0.33×
hashJoinPressure	F	5/15	67%	0.0202	0.0324	0.0718	0.0858	0.28×	1.60 ×	0.45×	0.38×
scaleCosts	M	15/15	0%	0.0691	0.0760	0.0856	0.2949	0.81×	1.10 ×	0.89×	0.26×
sumCost	F	6/15	60%	0.0304	0.0393	0.0160	0.0895	1.89 ×	1.29 ×	2.45 ×	0.44×
sumMemory	F	6/15	60%	0.0197	0.0286	0.0701	0.0881	0.28×	1.45 ×	0.41×	0.32×
sumRows	F	6/15	60%	0.0273	0.0300	0.0220	0.0869	1.24 ×	1.10×	1.36 ×	0.34×
Total				0.2746	0.3385	0.3891	1.248	0.71×	1.23 ×	0.87×	0.27×
Geomean								0.79×	1.30 ×	1.02×	0.31×

Table 10: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.6e+08 /2.2e+08	1.9e+08 /7.2e+08	2.9e+05 /1.4e+06	8.9e+04 /1.7e+05	3.1e+04/ 2.4e+04	2.6e+03 /1.1e+04	2.3e+06/ 9.2e+05
countJoins	F	1.0e+08/ 5.9e+07	7.4e+07/ 7.1e+07	6.1e+05/ 6.5e+04	4.0e+02 /4.9e+02	1.4e+05/ 2.4e+03	3.8e+02 /4.4e+02	2.7e+06/ 4.8e+04
filterSelectivitySkew	F	1.0e+08/ 6.0e+07	1.0e+08 /1.1e+08	5.0e+05/ 6.2e+04	2.6e+02/ 1.9e+02	1.2e+05/ 6.1e+03	2.4e+02/ 1.6e+02	2.7e+06/ 2.9e+05
hashJoinPressure	F	1.0e+08/ 6.2e+07	8.4e+07 /9.9e+07	7.3e+05/ 1.0e+05	2.9e+02 /3.4e+02	1.9e+05/ 7.4e+03	2.7e+02 /2.8e+02	2.8e+06/ 1.4e+05
scaleCosts	M	1.6e+08 /2.2e+08	2.2e+08 /7.5e+08	2.8e+05 /1.3e+06	7.5e+04 /1.8e+05	3.6e+04/ 2.8e+04	4.4e+03 /2.0e+04	2.5e+06/ 1.1e+06
sumCost	F	9.2e+07/ 7.7e+07	9.0e+07 /1.2e+08	2.5e+06/ 4.1e+05	4.8e+02/ 3.4e+02	2.4e+05/ 1.5e+04	3.6e+02/ 2.5e+02	2.3e+06/ 3.5e+05
sumMemory	F	9.9e+07/ 7.7e+07	9.0e+07 /1.2e+08	6.7e+05/ 2.1e+05	2.6e+02/ 2.1e+02	1.6e+05/ 1.3e+04	2.3e+02/ 1.5e+02	2.7e+06/ 2.9e+05
sumRows	F	1.3e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.8e+05/ 1.1e+05	1.8e+02 /2.5e+02	2.3e+04/ 7.5e+03	1.7e+02 /2.4e+02	2.4e+06/ 3.1e+05
Total		9.6e+08/8.9e+08	1.1e+09/2.2e+09	5.8e+06/3.7e+06	1.7e+05/3.5e+05	9.3e+05/1.0e+05	8.6e+03/3.2e+04	2.0e+07/3.4e+06

Table 11: Per-pass PAPI counters for **DBQuery** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.0e+08 /1.3e+09	3.2e+08 /2.4e+09	6.3e+05 /1.9e+07	9.3e+04 /6.8e+05	1.5e+04 /5.0e+04	2.9e+03 /4.4e+04	2.1e+06/ 9.9e+05
countJoins	F	1.4e+08 /4.3e+08	1.6e+08 /7.7e+08	1.9e+06/ 3.1e+05	3.0e+02 /8.8e+02	1.1e+05/ 1.3e+04	2.8e+02/ 2.0e+02	2.4e+06/ 8.0e+04
filterSelectivitySkew	F	1.4e+08 /4.4e+08	2.0e+08 /7.9e+08	1.4e+06/ 8.4e+05	2.4e+02 /1.3e+03	6.4e+04/ 1.7e+04	2.0e+02 /5.6e+02	2.4e+06/ 2.3e+05
hashJoinPressure	F	1.6e+08 /4.4e+08	1.8e+08 /7.7e+08	1.7e+06/ 6.4e+05	1.8e+02 /1.2e+03	1.1e+05/ 1.6e+04	1.7e+02 /2.7e+02	2.3e+06/ 1.6e+05
scaleCosts	M	2.0e+08 /1.3e+09	3.5e+08 /2.4e+09	5.4e+05 /2.3e+07	7.5e+04 /3.2e+05	1.7e+04 /6.2e+04	3.0e+03 /5.1e+04	2.3e+06/ 1.1e+06
sumCost	F	1.4e+08 /4.4e+08	1.8e+08 /7.8e+08	2.2e+06/ 1.2e+06	6.1e+02 /1.4e+03	1.1e+05/ 2.1e+04	4.7e+02 /8.2e+02	2.2e+06/ 3.5e+05
sumMemory	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.5e+06/ 1.1e+06	3.3e+02 /1.1e+03	7.0e+04/ 2.0e+04	3.4e+02 /6.2e+02	2.4e+06/ 3.3e+05
sumRows	F	1.5e+08 /4.4e+08	2.3e+08 /8.3e+08	1.5e+06/ 1.2e+06	2.6e+02 /1.0e+03	5.9e+04/ 2.1e+04	2.3e+02 /6.5e+02	2.3e+06/ 3.5e+05
Total		1.3e+09/5.2e+09	1.8e+09/9.5e+09	1.1e+07/4.7e+07	1.7e+05/1.0e+06	5.5e+05/2.2e+05	7.5e+03/9.8e+04	1.8e+07/3.6e+06

Table 12: Per-pass performance for **DecisionTree**, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); ± shows standard error of the mean across −iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	3.419	4.494	2.967	20.98	1.15 ×	1.31 ×	1.51 ×	0.21×
classify Depth	F	4/7	43%	0.0488	0.0641	0.0386	0.3010	1.26 ×	1.31 ×	1.66 ×	0.21×
classify tree	F	5/7	29%	0.0488	0.0641	0.0397	0.2979	1.23 ×	1.31 ×	1.62 ×	0.22×
countFeatureUses	F	3/7	57%	0.0522	0.1419	0.0316	0.3789	1.65 ×	2.72 ×	4.49 ×	0.37×
countLeaves	F	2/7	71%	0.0497	0.1409	0.0264	0.3716	1.88 ×	2.83 ×	5.33 ×	0.38×
countNodes	F	2/7	71%	0.0500	0.1456	0.0441	0.3699	1.14 ×	2.91 ×	3.30 ×	0.39×
countSmallLeaves	F	3/7	57%	0.0528	0.1364	0.0323	0.3715	1.64 ×	2.58 ×	4.23 ×	0.37×
inferenceCost	F	2/7	71%	0.0524	0.1390	0.0292	0.3706	1.79 ×	2.65 ×	4.76 ×	0.38×
sumImpurity	F	3/7	57%	0.0523	0.1395	0.0325	0.3813	1.61 ×	2.66 ×	4.30 ×	0.37×
sumPathLengths	F	3/7	57%	0.0550	0.1391	0.0338	0.3873	1.63 ×	2.53 ×	4.11 ×	0.36×
sumSamples	F	3/7	57%	0.0504	0.1355	0.0316	0.3838	1.59 ×	2.69 ×	4.29 ×	0.35×
treeDepth	F	2/7	71%	0.0529	0.1391	0.0285	0.3715	1.86 ×	2.63 ×	4.88 ×	0.37×
Total				3.984	5.879	3.335	24.96	1.19 ×	1.48 ×	1.76 ×	0.24×
Geomean								1.51 ×	2.25 ×	3.40 ×	0.32×

Table 13: Per-pass PAPI counters for **DecisionTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.7e+10/ 1.4e+10	3.3e+10 /5.9e+10	4.5e+08/ 2.7e+07	8.0e+04/ 4.7e+04	1.8e+06 /2.0e+06	8.0e+04/ 4.3e+04	9.2e+08/ 1.8e+07
classify Depth	F	2.5e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.5e+06/ 4.1e+05	1.0e+03/ 4.4e+02	2.2e+04 /2.8e+04	9.9e+02/ 4.1e+02	1.3e+07/ 3.3e+05
classify tree	F	2.5e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.5e+06/ 4.1e+05	1.0e+03/ 3.9e+02	2.1e+04 /2.8e+04	9.8e+02/ 3.4e+02	1.3e+07/ 3.4e+05
countFeatureUses	F	2.6e+08/ 1.6e+08	7.5e+08 /7.8e+08	7.6e+06/ 3.4e+05	8.1e+02/ 6.5e+02	3.7e+04/ 1.4e+04	8.0e+02/ 6.3e+02	1.4e+07/ 2.1e+06
countLeaves	F	2.5e+08/ 1.3e+08	6.0e+08 /6.1e+08	7.2e+06/ 7.6e+05	1.3e+03/ 7.6e+02	4.5e+04/ 1.7e+04	1.3e+03/ 7.3e+02	1.4e+07/ 2.3e+05
countNodes	F	2.5e+08/ 1.3e+08	6.1e+08 /6.2e+08	7.0e+06/ 7.6e+05	1.3e+03/ 1.0e+03	5.2e+04/ 1.7e+04	1.1e+03/ 8.1e+02	1.4e+07/ 2.3e+05
countSmallLeaves	F	2.7e+08/ 1.6e+08	7.1e+08 /8.4e+08	5.9e+06/ 8.9e+04	7.2e+02/ 6.9e+02	3.5e+04/ 1.4e+04	7.2e+02/ 6.7e+02	1.5e+07/ 2.0e+06
inferenceCost	F	2.7e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.0e+06/ 1.2e+05	1.2e+03/ 8.7e+02	1.8e+04/ 8.1e+03	1.2e+03/ 8.5e+02	1.4e+07/ 4.2e+05
sumImpurity	F	2.7e+08/ 1.6e+08	7.6e+08 /7.8e+08	7.8e+06/ 1.0e+06	1.4e+03/ 6.3e+02	3.6e+04/ 1.3e+04	1.4e+03/ 6.1e+02	1.4e+07/ 2.1e+06
sumPathLengths	F	2.8e+08/ 1.7e+08	8.0e+08 /9.0e+08	7.7e+06/ 4.6e+05	1.2e+03/ 7.3e+02	4.0e+04/ 1.4e+04	1.2e+03/ 7.0e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.6e+08/ 1.6e+08	6.4e+08 /7.7e+08	6.3e+06/ 2.4e+05	9.1e+02/ 6.3e+02	4.2e+04/ 1.4e+04	9.1e+02/ 6.0e+02	1.5e+07/ 2.1e+06
treeDepth	F	2.7e+08/ 1.4e+08	6.1e+08/ 6.1e+08	8.0e+06/ 5.9e+05	1.1e+03/ 6.7e+02	1.9e+04/ 1.5e+04	1.1e+03/ 6.2e+02	1.4e+07/ 2.8e+05
Total		2.0e+10/1.6e+10	4.0e+10/6.8e+10	5.3e+08/3.2e+07	9.2e+04/5.4e+04	2.2e+06/2.1e+06	9.1e+04/5.0e+04	1.1e+09/3.0e+07

Table 14: Per-pass PAPI counters for **DecisionTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	2.3e+10 /1.1e+11	5.1e+10 /1.9e+11	1.5e+07 /4.7e+07	2.7e+04 /1.1e+05	3.6e+05 /2.2e+06	2.7e+04/ 1.6e+04	8.4e+08/ 1.8e+07
classify Depth	F	3.2e+08 /1.5e+09	7.4e+08 /2.7e+09	2.4e+05 /6.3e+05	2.8e+02 /2.2e+03	5.0e+03 /3.0e+04	2.6e+02 /4.0e+02	1.2e+07/ 3.4e+05
classify tree	F	3.2e+08 /1.5e+09	7.4e+08 /2.7e+09	2.4e+05 /6.8e+05	2.8e+02 /2.2e+03	5.0e+03 /3.2e+04	2.6e+02 /5.7e+02	1.2e+07/ 3.3e+05
countFeatureUses	F	7.2e+08 /1.9e+09	1.3e+09 /3.1e+09	2.0e+06 /3.2e+06	4.8e+02 /2.7e+03	5.1e+03 /7.0e+04	4.5e+02 /1.2e+03	9.4e+06/ 1.2e+06
countLeaves	F	7.1e+08 /1.9e+09	1.2e+09 /2.9e+09	4.4e+06/ 6.7e+05	1.9e+03 /3.3e+03	9.3e+03 /3.2e+04	1.8e+03/ 6.5e+02	9.1e+06/ 3.8e+05
countNodes	F	7.3e+08 /1.9e+09	1.2e+09 /3.0e+09	5.7e+06/ 6.8e+05	1.0e+03 /2.5e+03	4.4e+04/ 3.2e+04	8.7e+02/ 4.1e+02	9.1e+06/ 3.8e+05
countSmallLeaves	F	6.9e+08 /1.9e+09	1.3e+09 /2.9e+09	1.1e+06 /3.4e+06	4.9e+02 /2.2e+03	3.8e+03 /7.3e+04	4.6e+02 /7.5e+02	9.2e+06/ 1.2e+06
inferenceCost	F	7.1e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.3e+05	5.9e+02 /3.4e+03	4.6e+03 /3.1e+04	5.5e+02 /5.9e+02	9.1e+06/ 3.8e+05
sumImpurity	F	7.1e+08 /1.9e+09	1.3e+09 /3.0e+09	2.2e+06 /3.5e+06	2.8e+02 /1.2e+03	5.5e+03 /7.5e+04	2.5e+02 /1.0e+03	9.4e+06/ 1.2e+06
sumPathLengths	F	7.1e+08 /2.0e+09	1.3e+09 /3.1e+09	1.5e+06 /3.6e+06	5.9e+02 /3.9e+03	4.1e+03 /7.9e+04	5.5e+02 /1.5e+03	9.1e+06/ 1.2e+06
sumSamples	F	6.9e+08 /2.0e+09	1.2e+09 /3.0e+09	1.1e+06 /3.6e+06	8.1e+02 /2.3e+03	4.5e+03 /7.7e+04	7.8e+02 /7.9e+02	9.2e+06/ 1.2e+06
treeDepth	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.2e+05	4.4e+02 /1.9e+03	4.9e+03 /3.1e+04	4.0e+02 /5.4e+02	9.1e+06/ 4.0e+05
Total		3.0e+10/1.3e+11	6.4e+10/2.2e+11	4.2e+07/6.9e+07	3.4e+04/1.4e+05	4.5e+05/2.8e+06	3.4e+04/2.4e+04	9.4e+08/2.6e+07

Table 15: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0364	0.0834	0.0205	0.1120	1.77 $\times$	2.29 $\times$	4.07 $\times$	0.74 $\times$
computeWidths	M	14/15	7%	0.0297	0.0305	0.0441	0.0878	0.67 $\times$	1.03 $\times$	0.69 $\times$	0.35 $\times$
count styled	F	3/15	80%	0.0345	0.0700	0.0117	0.0898	2.95 $\times$	2.03 $\times$	6.00 $\times$	0.78 $\times$
find max Bottom	F	5/15	67%	0.0345	0.0774	0.0181	0.0888	1.91 $\times$	2.24 $\times$	4.28 $\times$	0.87 $\times$
scaleLayout	M	15/15	0%	0.0287	0.0306	0.0364	0.0873	0.79 $\times$	1.07 $\times$	0.84 $\times$	0.35 $\times$
sumTextWidth	F	3/15	80%	0.0343	0.0702	0.0114	0.0881	3.00 $\times$	2.05 $\times$	6.15 $\times$	0.80 $\times$
Total				0.1980	0.3620	0.1421	0.5538	1.39 $\times$	1.83 $\times$	2.55 $\times$	0.65 $\times$
Geomean								1.59 $\times$	1.69 $\times$	2.68 $\times$	0.61 $\times$

Table 16: Per-pass PAPI counters for **DomTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.8e+08/ 9.7e+07	2.9e+08 /4.9e+08	2.5e+06/ 9.8e+05	9.0e+02/ 4.2e+02	1.6e+05/ 1.7e+04	7.4e+02/ 3.2e+02	1.4e+07/ 1.8e+06
computeWidths	M	6.6e+07 /1.4e+08	1.4e+08 /4.6e+08	1.2e+06 /1.5e+06	7.0e+04/ 6.4e+04	4.2e+03 /9.7e+03	2.2e+03 /1.6e+04	1.3e+06/ 4.5e+05
count styled	F	1.7e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.2e+06/ 2.4e+04	6.7e+02/ 1.4e+02	3.0e+05/ 5.0e+03	6.3e+02/ 1.3e+02	1.4e+07/ 7.1e+05
find max Bottom	F	1.7e+08/ 9.2e+07	2.5e+08 /4.1e+08	2.5e+06/ 8.7e+05	4.7e+02/ 1.6e+02	1.6e+05/ 9.4e+03	4.3e+02/ 1.4e+02	1.4e+07/ 1.5e+06
scaleLayout	M	5.9e+07 /9.9e+07	8.8e+07 /2.9e+08	1.1e+06 /2.3e+06	6.3e+04/ 5.1e+04	2.5e+04/ 9.0e+03	3.0e+03 /1.5e+04	1.4e+06/ 4.4e+05
sumTextWidth	F	1.7e+08/ 5.8e+07	2.0e+08 /2.4e+08	2.5e+06/ 1.8e+04	8.5e+02/ 1.9e+02	4.5e+05/ 4.3e+03	8.1e+02/ 1.8e+02	1.4e+07/ 7.2e+05
Total		8.3e+08/5.4e+08	1.2e+09/2.2e+09	1.2e+07/5.7e+06	1.4e+05/1.2e+05	1.1e+06/5.5e+04	7.8e+03/3.2e+04	5.7e+07/5.6e+06

Table 17: Per-pass PAPI counters for **DomTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	4.2e+08 /4.7e+08	4.4e+08 /1.8e+09	1.3e+07/ 2.8e+06	5.5e+02 /1.5e+03	4.3e+05/ 5.1e+04	4.4e+02 /1.4e+03	9.1e+06/ 1.3e+06
computeWidths	M	6.8e+07 /3.5e+08	1.5e+08 /9.0e+08	1.4e+06 /3.3e+06	4.6e+04 /2.6e+05	4.1e+03 /1.7e+04	1.0e+03 /2.6e+04	1.2e+06/ 4.7e+05
count styled	F	3.5e+08 /4.6e+08	4.3e+08 /1.7e+09	1.0e+07/ 1.2e+06	2.5e+02 /5.4e+02	3.6e+04 /3.7e+04	2.4e+02 /5.0e+02	1.0e+07/ 4.7e+05
find max Bottom	F	3.8e+08 /4.5e+08	4.6e+08 /1.8e+09	1.2e+07/ 1.3e+06	3.2e+02 /1.0e+03	2.8e+05/ 2.7e+04	3.0e+02 /1.0e+03	9.7e+06/ 1.0e+06
scaleLayout	M	6.8e+07 /3.5e+08	1.1e+08 /7.6e+08	9.9e+05 /4.1e+06	5.4e+04 /9.8e+04	8.8e+03 /1.2e+04	1.3e+03 /2.5e+04	1.2e+06/ 4.7e+05
sumTextWidth	F	3.6e+08 /4.5e+08	3.9e+08 /1.7e+09	1.1e+07/ 1.3e+06	3.0e+02 /9.8e+02	8.8e+04/ 3.9e+04	2.8e+02 /8.1e+02	1.0e+07/ 4.5e+05
Total		1.6e+09/2.5e+09	2.0e+09/8.6e+09	4.8e+07/1.4e+07	1.0e+05/3.6e+05	8.4e+05/1.8e+05	3.6e+03/5.4e+04	4.2e+07/4.2e+06

Table 18: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	13/17	24%	0.0288	0.0474	0.0476	0.1188	0.61 $\times$	1.65 $\times$	1.00 $\times$	0.40 $\times$
countInRange tight_box	F	13/17	24%	0.0282	0.0433	0.0240	0.1241	1.18 $\times$	1.53 $\times$	1.80 $\times$	0.35 $\times$
photonMappingBenchmark	F	12/17	29%	0.0431	0.0507	0.0415	0.1331	1.04 $\times$	1.18 $\times$	1.22 $\times$	0.38 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0254	0.0419	0.0163	0.1151	1.55 $\times$	1.65 $\times$	2.56 $\times$	0.36 $\times$
sumMassInRange	F	14/17	18%	0.0277	0.0436	0.0200	0.1245	1.38 $\times$	1.58 $\times$	2.18 $\times$	0.35 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0292	0.0423	0.0210	0.1137	1.39 $\times$	1.45 $\times$	2.01 $\times$	0.37 $\times$
Total				0.1823	0.2692	0.1705	0.7294	1.07 $\times$	1.48 $\times$	1.58 $\times$	0.37 $\times$
Geomean								1.14 $\times$	1.50 $\times$	1.71 $\times$	0.37 $\times$

Table 19: Per-pass PAPI counters for **KDTree** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.5e+08/ 1.0e+08	4.7e+08 /6.3e+08	2.7e+06/ 2.0e+06	5.0e+02 /5.9e+02	4.2e+03 /1.8e+04	3.7e+02 /4.6e+02	7.8e+06/ 1.5e+06
countInRange tight_box	F	1.4e+08/ 9.8e+07	4.0e+08 /5.5e+08	2.0e+06 /2.1e+06	1.8e+02/ 1.7e+02	3.8e+03 /1.9e+04	1.5e+02 /1.7e+02	7.8e+06/ 1.5e+06
photonMappingBenchmark	F	2.2e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.1e+06 /2.5e+06	3.5e+02 /5.8e+02	6.1e+03 /1.8e+04	3.3e+02 /5.6e+02	5.6e+06/ 1.5e+06
pointCloudNeighborhood	F	1.3e+08/ 8.3e+07	3.3e+08 /4.8e+08	2.5e+06/ 1.3e+06	1.5e+02 /1.6e+02	3.7e+03 /1.7e+04	1.3e+02 /1.6e+02	7.8e+06/ 1.4e+06
sumMassInRange	F	1.4e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.4e+06 /3.5e+06	2.3e+02/ 2.1e+02	3.8e+03 /2.3e+04	2.1e+02 /2.2e+02	7.7e+06/ 1.8e+06
twoPointCorrelation bin_8_16	F	1.5e+08/ 1.1e+08	4.8e+08 /6.1e+08	2.1e+06/ 6.4e+05	2.1e+02/ 2.0e+02	3.6e+03 /1.8e+04	1.9e+02 /2.0e+02	7.7e+06/ 1.3e+06
Total		9.2e+08/7.0e+08	3.1e+09/4.0e+09	1.4e+07/1.2e+07	1.6e+03/1.9e+03	2.5e+04/1.1e+05	1.4e+03/1.8e+03	4.4e+07/8.9e+06

Table 20: Per-pass PAPI counters for **KDTree** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.1e+08 /6.0e+08	4.9e+08 /1.3e+09	5.1e+06 /5.3e+06	4.9e+02 /2.1e+03	4.4e+04/ 2.5e+04	3.3e+02 /1.9e+03	5.4e+06/ 1.5e+06
countInRange tight_box	F	2.2e+08 /6.3e+08	4.5e+08 /1.3e+09	5.1e+06 /5.8e+06	1.8e+02 /2.3e+03	5.3e+03 /2.2e+04	1.7e+02 /2.3e+03	5.6e+06/ 1.6e+06
photonMappingBenchmark	F	2.6e+08 /6.7e+08	1.1e+09 /2.0e+09	2.9e+06 /5.9e+06	5.5e+02 /1.8e+03	7.5e+03 /2.2e+04	5.1e+02 /1.7e+03	5.1e+06/ 1.5e+06
pointCloudNeighborhood	F	2.1e+08 /5.8e+08	4.1e+08 /1.3e+09	5.1e+06/ 4.8e+06	2.9e+02 /1.3e+03	5.9e+03 /1.9e+04	2.7e+02 /1.3e+03	5.8e+06/ 1.4e+06
sumMassInRange	F	2.2e+08 /6.3e+08	4.7e+08 /1.3e+09	5.1e+06 /6.3e+06	2.8e+02 /1.6e+03	4.9e+03 /2.5e+04	2.6e+02 /1.6e+03	5.6e+06/ 1.6e+06
twoPointCorrelation bin_8_16	F	2.1e+08 /5.8e+08	6.1e+08 /1.4e+09	4.7e+06/ 3.7e+06	3.0e+02 /1.4e+03	6.3e+03 /1.7e+04	2.8e+02 /1.4e+03	5.8e+06/ 1.3e+06
Total		1.3e+09/3.7e+09	3.5e+09/8.4e+09	2.8e+07/3.2e+07	2.1e+03/1.1e+04	7.4e+04/1.3e+05	1.8e+03/1.0e+04	3.3e+07/8.9e+06

Table 21: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0301	0.1340	0.0063	<i>OOM</i>	4.77 $\times$	4.45 $\times$	21.23 $\times$	–
Total				0.0301	0.1340	0.0063	–	4.77 $\times$	4.45 $\times$	21.23 $\times$	–
Geomean								4.77 $\times$	4.45 $\times$	21.23 $\times$	–

Table 22: Per-pass PAPI counters for **LinearListReduction** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.5e+08/ 2.6e+07	9.0e+07 /1.0e+08	1.5e+06/ 1.2e+06	7.8e+02/ 3.8e+02	2.8e+05/ 6.7e+04	6.7e+02/ 2.0e+02	1.2e+07/ 1.1e+06
Total		1.5e+08/2.6e+07	9.0e+07/1.0e+08	1.5e+06/1.2e+06	7.8e+02/3.8e+02	2.8e+05/6.7e+04	6.7e+02/2.0e+02	1.2e+07/1.1e+06

Table 23: Per-pass PAPI counters for **LinearListReduction** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	6.7e+08/–	2.5e+08/–	4.0e+06/–	9.3e+02/–	4.1e+05/–	7.2e+02/–	2.7e+07/–
Total		6.7e+08/–	2.5e+08/–	4.0e+06/–	9.3e+02/–	4.1e+05/–	7.2e+02/–	2.7e+07/–

Table 24: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.2772	<i>OOM</i>	0.3409	<i>OOM</i>	0.81 $\times$	–	–	–
length List	F	1/2	50%	0.0435	<i>OOM</i>	0.0216	<i>OOM</i>	2.01 $\times$	–	–	–
sumList	F	2/2	0%	0.0441	<i>OOM</i>	0.0449	<i>OOM</i>	0.98 $\times$	–	–	–
sumList tail recursive	F	2/2	0%	0.0442	<i>OOM</i>	0.0447	<i>OOM</i>	0.99 $\times$	–	–	–
Total				0.4091	–	0.4521	–	0.90 $\times$	–	–	–
Geomean								1.12 $\times$	–	–	–

Table 25: Per-pass PAPI counters for **List** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.8e+08	1.9e+09 /2.8e+09	8.1e+06/ 1.6e+06	3.4e+05 /6.1e+05	1.7e+04 /6.0e+04	5.1e+03 /1.3e+04	9.5e+06/ 7.7e+06
length List	F	2.2e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.1e+05/ 8.9e+04	1.6e+02/ 8.7e+01	2.8e+03 /1.4e+04	1.4e+02/ 7.4e+01	1.3e+07/ 1.2e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	7.0e+05/ 5.3e+05	1.9e+02/ 8.3e+01	3.1e+03 /6.9e+04	1.7e+02/ 6.9e+01	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	5.6e+05 /5.6e+05	2.1e+02/ 2.0e+02	2.9e+03 /5.8e+04	1.9e+02/ 1.9e+02	1.3e+07/ 1.2e+07
Total		1.3e+09/1.5e+09	4.5e+09/5.6e+09	9.5e+06/2.8e+06	3.4e+05/6.1e+05	2.5e+04/2.0e+05	5.6e+03/1.3e+04	4.8e+07/3.3e+07

Table 26: Per-pass PAPI counters for **List** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		–/–	–/–	–/–	–/–	–/–	–/–	–/–

Table 27: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0952	0.2162	0.1025	0.1608	0.93 $\times$	2.27 $\times$	2.11 $\times$	1.34 $\times$
sumTree	F	3/3	0%	0.0092	0.0401	0.0113	0.0808	0.82 $\times$	4.34 $\times$	3.55 $\times$	0.50 $\times$
sumTree TailRec	F	3/3	0%	0.0087	0.0401	0.0112	0.0808	0.78 $\times$	4.59 $\times$	3.59 $\times$	0.50 $\times$
Total				0.1132	0.2964	0.1249	0.3224	0.91 $\times$	2.62 $\times$	2.37 $\times$	0.92 $\times$
Geomean								0.84 $\times$	3.56 $\times$	3.00 $\times$	0.69 $\times$

Table 28: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	2.5e+05 /8.3e+05	3.5e+04 /6.6e+04	2.9e+03 /1.1e+04	1.9e+03/ 1.7e+03	4.6e+05/ 4.2e+05
sumTree	F	4.7e+07 /5.7e+07	2.2e+08 /2.6e+08	8.0e+03 /3.3e+04	2.8e+02 /4.8e+02	7.6e+02 /4.0e+03	2.3e+02 /4.5e+02	8.7e+05/ 7.0e+05
sumTree TailRec	F	4.4e+07 /5.6e+07	1.9e+08 /2.2e+08	6.2e+03 /2.1e+04	3.0e+02 /7.1e+02	7.8e+02 /3.9e+03	2.8e+02 /6.9e+02	8.7e+05/ 7.0e+05
Total		2.6e+08/2.6e+08	9.1e+08/1.1e+09	2.6e+05/8.8e+05	3.5e+04/6.7e+04	4.5e+03/1.9e+04	2.4e+03/2.9e+03	2.2e+06/1.8e+06

Table 29: Per-pass PAPI counters for `MonoTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.4e+08 /5.0e+08	7.1e+08 /1.2e+09	3.5e+05 /9.8e+05	4.6e+04 /7.4e+04	4.8e+03 /1.5e+04	2.9e+03/ 8.5e+02	4.0e+05 /4.3e+05
sumTree	F	2.0e+08 /4.1e+08	4.3e+08 /6.3e+08	6.1e+04 /9.5e+05	2.8e+02 /4.3e+02	1.1e+03 /1.5e+04	2.6e+02 /3.8e+02	3.6e+05 /4.0e+05
sumTree TailRec	F	2.0e+08 /4.1e+08	4.1e+08 /6.4e+08	4.5e+04 /9.3e+05	2.4e+02/ 1.8e+02	1.0e+03 /1.4e+04	2.2e+02/ 1.5e+02	3.5e+05 /4.0e+05
Total		6.5e+08/1.3e+09	1.6e+09/2.5e+09	4.5e+05/2.9e+06	4.7e+04/7.4e+04	6.9e+03/4.3e+04	3.4e+03/1.4e+03	1.1e+06/1.2e+06

Table 30: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
countLargeItems	F	3/5	40%	0.0133	0.0443	0.0109	0.0468	1.22 $\times$	3.34 $\times$	4.06 $\times$	0.95 $\times$
countMarked	F	3/5	40%	0.0134	0.0448	0.0147	0.0470	0.91 $\times$	3.36 $\times$	3.05 $\times$	0.95 $\times$
countSurvivors	F	4/5	20%	0.0142	0.0447	0.0130	0.0468	1.10 $\times$	3.14 $\times$	3.45 $\times$	0.95 $\times$
deadBytes	F	4/5	20%	0.0139	0.0440	0.0125	0.0470	1.12 $\times$	3.16 $\times$	3.53 $\times$	0.94 $\times$
liveBytes	F	4/5	20%	0.0138	0.0432	0.0125	0.0472	1.11 $\times$	3.12 $\times$	3.46 $\times$	0.91 $\times$
sumObjIds	F	3/5	40%	0.0133	0.0432	0.0103	0.0472	1.29 $\times$	3.25 $\times$	4.18 $\times$	0.92 $\times$
sweepUnmarked	M	5/5	0%	0.0730	0.0894	0.0810	0.1872	0.90 $\times$	1.22 $\times$	1.10 $\times$	0.48 $\times$
totalHeapSize	F	3/5	40%	0.0133	0.1106	0.0205	0.0500	0.65 $\times$	8.29 $\times$	5.40 $\times$	2.21 $\times$
touchHotObjects	M	5/5	0%	0.0722	0.0881	0.0828	0.1913	0.87 $\times$	1.22 $\times$	1.06 $\times$	0.46 $\times$
Total				0.2405	0.5523	0.2581	0.7104	0.93 $\times$	2.30 $\times$	2.14 $\times$	0.78 $\times$
Geomean								1.00 $\times$	2.89 $\times$	2.89 $\times$	0.88 $\times$

Table 31: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.7e+07/ 5.3e+07	2.6e+08 /2.8e+08	1.2e+06/ 4.6e+04	5.8e+02/ 3.3e+02	3.4e+03 /3.6e+03	5.5e+02/ 3.1e+02	2.9e+06/ 7.2e+05
countMarked	F	6.8e+07/ 5.4e+07	2.6e+08 /2.8e+08	1.2e+06/ 4.5e+04	3.9e+02/ 3.4e+02	3.5e+03 /4.0e+03	3.2e+02/ 3.2e+02	3.0e+06/ 6.7e+05
countSurvivors	F	7.2e+07/ 6.6e+07	3.1e+08 /3.8e+08	9.5e+05/ 6.2e+04	3.9e+02/ 2.6e+02	2.4e+03 /5.5e+03	3.6e+02/ 2.5e+02	2.8e+06/ 1.3e+06
deadBytes	F	7.1e+07/ 6.3e+07	2.7e+08 /3.5e+08	7.0e+05/ 5.3e+04	2.7e+02 /3.7e+02	2.5e+03 /5.7e+03	2.5e+02 /3.4e+02	2.8e+06/ 1.3e+06
liveBytes	F	7.0e+07/ 6.3e+07	2.7e+08 /3.5e+08	9.3e+05/ 5.1e+04	2.5e+02 /3.3e+02	2.4e+03 /5.8e+03	2.3e+02 /3.1e+02	2.8e+06/ 1.3e+06
sumObjIds	F	6.7e+07/ 5.2e+07	2.5e+08 /2.7e+08	1.2e+06/ 8.7e+04	2.8e+02/ 2.7e+02	2.8e+03 /4.5e+03	2.6e+02/ 2.5e+02	2.8e+06/ 7.0e+05
sweepUnmarked	M	1.9e+08 /2.4e+08	5.5e+08 /8.8e+08	1.1e+06/ 4.8e+05	6.9e+04 /1.1e+05	6.1e+03 /1.1e+04	5.0e+03 /7.5e+03	2.0e+06/ 1.0e+06
totalHeapSize	F	6.7e+07/ 5.4e+07	2.5e+08 /2.7e+08	1.2e+06/ 7.4e+05	5.8e+02 /6.3e+02	2.7e+03 /1.6e+04	4.8e+02 /5.3e+02	2.9e+06/ 5.5e+05
touchHotObjects	M	1.9e+08 /2.4e+08	5.7e+08 /9.0e+08	1.2e+06/ 5.0e+05	9.4e+04 /1.3e+05	5.1e+03 /1.1e+04	4.9e+03 /6.9e+03	1.9e+06/ 1.0e+06
Total		8.6e+08/8.8e+08	3.0e+09/4.0e+09	9.6e+06/2.1e+06	1.7e+05/2.4e+05	3.1e+04/6.7e+04	1.2e+04/1.7e+04	2.4e+07/8.5e+06

Table 32: Per-pass PAPI counters for `ObjectGraph` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.2e+08 /2.4e+08	4.6e+08 /7.9e+08	5.6e+05/ 4.9e+05	3.9e+02/ 1.8e+02	4.8e+03 /8.1e+03	3.6e+02/ 1.4e+02	1.8e+06/ 4.2e+05
countMarked	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.6e+05/ 4.9e+05	1.8e+02 /7.8e+02	5.0e+03 /9.9e+03	1.6e+02 /7.4e+02	1.8e+06/ 4.4e+05
countSurvivors	F	2.3e+08 /2.4e+08	5.5e+08 /8.2e+08	1.7e+05 /4.1e+05	2.8e+02 /4.3e+02	1.8e+03 /6.6e+03	2.6e+02 /4.2e+02	1.8e+06/ 7.4e+05
deadBytes	F	2.2e+08 /2.4e+08	4.9e+08 /8.1e+08	3.9e+05 /4.2e+05	2.8e+02 /3.5e+02	3.7e+03 /6.7e+03	2.6e+02 /3.4e+02	1.8e+06/ 7.4e+05
liveBytes	F	2.2e+08 /2.4e+08	5.0e+08 /8.1e+08	1.6e+05 /4.1e+05	1.9e+02 /3.8e+02	2.1e+03 /6.7e+03	1.6e+02 /3.6e+02	1.8e+06/ 7.5e+05
sumObjIds	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.5e+05/ 5.1e+05	4.2e+02/ 2.2e+02	4.9e+03 /9.3e+03	4.0e+02/ 2.0e+02	1.8e+06/ 4.2e+05
sweepUnmarked	M	2.7e+08 /7.7e+08	7.6e+08 /2.1e+09	1.9e+05 /2.4e+06	1.0e+05 /1.4e+05	4.1e+03 /1.8e+04	2.0e+03 /5.7e+03	1.8e+06/ 9.3e+05
totalHeapSize	F	2.3e+08 /2.4e+08	4.4e+08 /7.8e+08	2.1e+06/ 9.8e+05	9.6e+02/ 7.8e+02	3.4e+04/ 1.9e+04	6.4e+02/ 6.4e+02	1.3e+06/ 3.8e+05
touchHotObjects	M	2.7e+08 /7.9e+08	7.7e+08 /2.2e+09	2.4e+05 /1.8e+06	1.1e+05 /1.2e+05	3.5e+03 /2.0e+04	1.7e+03 /5.4e+03	1.7e+06/ 9.3e+05
Total		2.1e+09/3.2e+09	4.9e+09/9.9e+09	4.9e+06/7.9e+06	2.1e+05/2.6e+05	6.5e+04/1.0e+05	5.9e+03/1.4e+04	1.6e+07/5.8e+06

Table 33: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
addConstPW	M	8/8	0%	0.1194	0.1308	0.1344	0.4022	0.89 $\times$	1.09 $\times$	0.97 $\times$	0.33 $\times$
autorefineMaxLevel	F	4/8	50%	0.0204	0.0448	0.0357	0.2388	0.57 $\times$	2.20 $\times$	1.26 $\times$	0.19 $\times$
compressMass	F	3/8	62%	0.0197	0.0452	0.0637	0.2380	0.31 $\times$	2.30 $\times$	0.71 $\times$	0.19 $\times$
diffPW	M	8/8	0%	0.1197	0.1322	0.1326	0.3995	0.90 $\times$	1.10 $\times$	1.00 $\times$	0.33 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0213	0.0468	0.0184	0.2414	1.15 $\times$	2.20 $\times$	2.54 $\times$	0.19 $\times$
norm2Estimate	F	5/8	38%	0.0253	0.0641	0.0313	0.2308	0.81 $\times$	2.53 $\times$	2.05 $\times$	0.28 $\times$
pmapCutHistogram	F	4/8	50%	0.0317	0.0555	0.0127	0.2456	2.48 $\times$	1.75 $\times$	4.36 $\times$	0.23 $\times$
truncateTolViolations	F	3/8	62%	0.0198	0.0461	0.0703	0.2050	0.28 $\times$	2.32 $\times$	0.66 $\times$	0.22 $\times$
Total				0.3773	0.5654	0.4990	2.201	0.76 $\times$	1.50 $\times$	1.13 $\times$	0.26 $\times$
Geomean								0.74 $\times$	1.85 $\times$	1.38 $\times$	0.24 $\times$

Table 34: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.7e+08 /3.6e+08	5.7e+08 /1.2e+09	1.5e+06 /2.1e+06	2.6e+05 /2.6e+05	1.4e+04 /2.4e+04	5.7e+03 /2.1e+04	4.6e+06/ 1.7e+06
autorefineMaxLevel	F	1.0e+08/ 9.8e+07	2.5e+08 /3.7e+08	2.4e+06/ 2.5e+05	7.0e+02/ 1.9e+02	5.8e+03 /6.6e+03	7.1e+02/ 1.6e+02	6.2e+06/ 7.5e+05
compressMass	F	1.0e+08/ 5.6e+07	2.5e+08 /2.8e+08	2.0e+06/ 3.6e+04	7.5e+02 /1.1e+03	2.0e+04/ 5.6e+03	7.4e+02 /9.4e+02	6.4e+06/ 4.7e+05
diffPW	M	2.8e+08 /3.4e+08	5.9e+08 /1.3e+09	8.1e+05 /2.1e+06	1.6e+05 /2.0e+05	9.2e+03 /2.4e+04	9.9e+03 /2.5e+04	4.2e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 9.4e+07	3.3e+08 /4.9e+08	2.3e+06/ 4.3e+05	4.5e+02/ 2.9e+02	1.4e+04/ 9.3e+03	4.4e+02/ 2.7e+02	6.3e+06/ 1.6e+06
norm2Estimate	F	1.3e+08/ 8.8e+07	3.4e+08 /4.4e+08	1.4e+06/ 5.2e+05	9.2e+02/ 8.6e+02	1.2e+04/ 1.1e+04	8.1e+02/ 7.5e+02	6.2e+06/ 1.1e+06
pmapCutHistogram	F	1.6e+08/ 6.4e+07	2.8e+08 /3.7e+08	1.7e+06/ 5.7e+04	1.2e+03/ 5.0e+02	1.0e+05/ 7.6e+03	1.2e+03/ 4.8e+02	4.9e+06/ 1.3e+06
truncateTolViolations	F	1.0e+08/ 5.6e+07	2.5e+08 /2.9e+08	2.0e+06/ 5.0e+04	8.1e+02 /1.0e+03	1.5e+04/ 5.2e+03	7.8e+02 /8.8e+02	6.4e+06/ 4.8e+05
Total		1.3e+09/1.2e+09	2.8e+09/4.7e+09	1.4e+07/5.6e+06	4.2e+05/4.7e+05	1.9e+05/9.2e+04	2.0e+04/4.9e+04	4.5e+07/9.2e+06

Table 35: Per-pass PAPI counters for `PiecewiseFunctions` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
addConstPW	M	3.3e+08 /1.7e+09	7.8e+08 /3.3e+09	3.2e+05 /5.1e+06	1.3e+05 /3.7e+05	7.8e+03 /4.1e+04	5.2e+03 /2.5e+04	4.2e+06/ 1.9e+06
autorefineMaxLevel	F	2.3e+08 /1.2e+09	4.9e+08 /1.3e+09	2.9e+05 /2.1e+06	3.2e+02 /1.5e+03	1.5e+03 /4.0e+04	3.0e+02 /5.6e+02	4.2e+06/ 7.0e+05
compressMass	F	2.3e+08 /1.2e+09	4.7e+08 /1.2e+09	3.1e+05 /1.3e+06	3.4e+02 /2.0e+03	1.7e+03 /3.0e+04	3.2e+02 /4.3e+02	4.1e+06/ 4.0e+05
diffPW	M	3.4e+08 /1.7e+09	8.1e+08 /3.3e+09	3.1e+05 /5.2e+06	1.2e+05 /3.1e+05	4.8e+03 /4.1e+04	5.7e+03 /1.9e+04	4.1e+06/ 1.8e+06
lbDeuxLoadProxy	F	2.4e+08 /1.2e+09	5.3e+08 /1.3e+09	1.6e+05 /2.6e+06	2.0e+02 /5.0e+02	1.7e+03 /4.2e+04	1.7e+02 /4.3e+02	4.1e+06/ 1.0e+06
norm2Estimate	F	2.9e+08 /1.0e+09	5.1e+08 /1.3e+09	1.9e+06 /2.3e+06	4.7e+02 /1.3e+03	3.7e+04 /3.9e+04	3.8e+02 /9.4e+02	3.8e+06/ 9.9e+05
pmapCutHistogram	F	2.8e+08 /1.2e+09	5.0e+08 /1.3e+09	7.6e+05 /2.0e+06	3.4e+02 /2.3e+03	1.0e+04 /3.7e+04	3.0e+02 /6.1e+02	3.9e+06/ 7.1e+05
truncateTolViolations	F	2.3e+08 /1.0e+09	4.4e+08 /1.3e+09	3.4e+05 /1.3e+06	4.7e+02 /1.6e+03	3.8e+03 /2.7e+04	4.1e+02 /4.7e+02	4.0e+06/ 4.1e+05
Total		2.2e+09/1.0e+10	4.5e+09/1.4e+10	4.4e+06/2.2e+07	2.6e+05/7.0e+05	6.8e+04/3.0e+05	1.3e+04/4.8e+04	3.2e+07/7.9e+06

Table 36: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add 1 tree	M	5/5	0%	0.0836	0.1573	0.0882	0.3757	0.95 $\times$	1.88$\times$	1.78$\times$	0.42 $\times$
sum tree	F	5/5	0%	0.0290	0.0500	0.0277	0.2730	1.05 $\times$	1.73$\times$	1.81$\times$	0.18 $\times$
Total				0.1126	0.2073	0.1159	0.6487	0.97 $\times$	1.84$\times$	1.79$\times$	0.32 $\times$
Geomean								1.00 $\times$	1.80$\times$	1.80$\times$	0.28 $\times$

Table 37: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	3.5e+05 /1.2e+06	6.6e+04 /1.3e+05	4.3e+03 /1.2e+04	2.7e+03/ 2.6e+03	1.6e+06/ 1.1e+06
sum tree	F	1.5e+08/ 1.4e+08	4.6e+08 /5.4e+08	3.7e+04 /3.1e+05	1.5e+02/ 1.2e+02	7.9e+02 /8.9e+03	1.1e+02/ 9.5e+01	1.8e+06/ 1.2e+06
Total		4.2e+08/3.7e+08	1.2e+09/1.5e+09	3.9e+05/1.5e+06	6.6e+04/1.3e+05	5.1e+03/2.1e+04	2.8e+03/2.7e+03	3.4e+06/2.3e+06

Table 38: Per-pass PAPI counters for `TernaryTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.2e+08 /1.6e+09	9.5e+08 /2.4e+09	2.1e+05 /2.4e+06	8.5e+04 /1.8e+05	3.5e+03 /4.4e+04	2.0e+03 /4.9e+03	1.3e+06/ 9.6e+05
sum tree	F	2.5e+08 /1.4e+09	5.8e+08 /1.0e+09	8.2e+04 /2.3e+06	1.8e+02 /1.3e+03	1.6e+03 /4.0e+04	1.5e+02 /3.9e+02	1.4e+06/ 9.6e+05
Total		5.8e+08/3.0e+09	1.5e+09/3.4e+09	2.9e+05/4.6e+06	8.5e+04/1.8e+05	5.1e+03/8.4e+04	2.1e+03/5.3e+03	2.7e+06/1.9e+06

Table 39: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
autocompleteTopKProxy	F	5/10	50%	0.0144	0.0268	0.0098	0.1075	1.47$\times$	1.86$\times$	2.73$\times$	0.25 $\times$
countLazyNodes	F	4/10	60%	0.0123	0.0271	0.0071	0.1086	1.75$\times$	2.20$\times$	3.84$\times$	0.25 $\times$
countTerminals	F	3/10	70%	0.0118	0.0251	0.0103	0.1010	1.14$\times$	2.13$\times$	2.43$\times$	0.25 $\times$
decayTrieStats	M	10/10	0%	0.0744	0.0799	0.0871	0.2373	0.85 $\times$	1.07 $\times$	0.92 $\times$	0.34 $\times$
resetTraversalState	M	10/10	0%	0.0754	0.0820	0.0823	0.2318	0.92 $\times$	1.09 $\times$	1.00 $\times$	0.35 $\times$
sumPrefixFreq	F	3/10	70%	0.0120	0.0451	0.0124	0.1070	0.97 $\times$	3.75$\times$	3.64$\times$	0.42 $\times$
sumSubtreeHints	F	3/10	70%	0.0116	0.0255	0.0085	0.1039	1.36$\times$	2.19$\times$	2.98$\times$	0.24 $\times$
Total				0.2120	0.3114	0.2175	0.9972	0.97 $\times$	1.47$\times$	1.43$\times$	0.31 $\times$
Geomean								1.17$\times$	1.88$\times$	2.21$\times$	0.29 $\times$

Table 40: Per-pass PAPI counters for `Trie` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	7.3e+07/ 4.4e+07	1.3e+08 /2.1e+08	1.5e+06 /1.7e+06	1.5e+02/ 1.2e+02	5.3e+04/ 3.2e+04	1.3e+02/ 1.1e+02	3.8e+06/ 7.6e+05
countLazyNodes	F	6.3e+07/ 3.6e+07	1.3e+08 /1.6e+08	1.3e+06/ 6.8e+05	2.1e+02/ 1.6e+02	1.5e+04/ 1.4e+04	1.8e+02/ 1.5e+02	4.1e+06/ 6.4e+05
countTerminals	F	5.9e+07/ 3.1e+07	1.0e+08 /1.2e+08	1.2e+06/ 6.4e+05	3.5e+02/ 1.8e+02	2.6e+04/ 1.3e+04	3.2e+02/ 1.5e+02	4.1e+06/ 2.5e+05
decayTrieStats	M	1.6e+08 /2.1e+08	3.5e+08 /7.9e+08	7.4e+05 /1.4e+06	9.4e+04 /1.1e+05	6.5e+03 /1.9e+04	3.1e+03 /2.3e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	1.6e+08 /2.0e+08	2.8e+08 /7.0e+08	1.0e+06/ 8.8e+05	8.8e+04 /1.1e+05	8.5e+03 /1.4e+04	2.5e+03 /1.7e+04	3.3e+06/ 9.1e+05
sumPrefixFreq	F	6.1e+07/ 3.0e+07	1.1e+08 /1.2e+08	1.3e+06/ 6.4e+05	5.8e+02/ 4.2e+02	2.7e+04/ 1.3e+04	4.5e+02/ 3.0e+02	4.1e+06/ 2.3e+05
sumSubtreeHints	F	5.9e+07/ 3.1e+07	1.1e+08 /1.2e+08	1.4e+06/ 6.4e+05	3.1e+02/ 1.3e+02	2.6e+04/ 1.3e+04	2.8e+02/ 1.2e+02	4.0e+06/ 2.8e+05
Total		6.4e+08/5.8e+08	1.2e+09/2.2e+09	8.5e+06/6.5e+06	1.8e+05/2.2e+05	1.6e+05/1.2e+05	6.9e+03/4.1e+04	2.6e+07/4.3e+06

Table 41: Per-pass PAPI counters for **Trie** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
autocompleteTopKProxy	F	1.4e+08 /5.5e+08	2.3e+08 /7.5e+08	1.0e+06 /1.7e+06	2.2e+02 /9.4e+02	3.0e+03 /3.3e+04	1.9e+02 /7.2e+02	2.8e+06/ 4.5e+05
countLazyNodes	F	1.4e+08 /5.5e+08	2.2e+08 /7.6e+08	2.2e+06/ 1.1e+06	1.4e+02 /7.4e+02	2.7e+03 /2.2e+04	1.3e+02 /2.5e+02	2.7e+06/ 3.5e+05
countTerminals	F	1.3e+08 /5.1e+08	2.0e+08 /7.3e+08	1.2e+06/ 7.0e+05	1.9e+02 /1.1e+03	1.6e+03 /1.7e+04	1.7e+02 /2.1e+02	2.9e+06/ 1.9e+05
decayTrieStats	M	1.9e+08 /9.8e+08	4.5e+08 /2.1e+09	3.2e+05 /5.2e+06	8.5e+04 /3.9e+05	4.0e+03 /3.2e+04	2.9e+03 /3.8e+04	2.9e+06/ 1.2e+06
resetTraversalState	M	2.0e+08 /9.6e+08	3.7e+08 /2.0e+09	3.0e+05 /3.3e+06	1.0e+05 /1.6e+05	4.6e+03 /2.7e+04	2.3e+03 /3.0e+04	2.9e+06/ 9.6e+05
sumPrefixFreq	F	1.5e+08 /5.3e+08	2.1e+08 /7.3e+08	4.0e+06/ 7.0e+05	5.0e+02 /1.4e+03	8.1e+04/ 1.7e+04	3.8e+02/ 3.7e+02	2.4e+06/ 1.9e+05
sumSubtreeHints	F	1.3e+08 /5.3e+08	2.1e+08 /7.3e+08	1.3e+06/ 7.0e+05	1.8e+02 /9.2e+02	1.3e+03 /1.7e+04	1.6e+02 /2.3e+02	2.9e+06/ 1.9e+05
Total		1.1e+09/4.6e+09	1.9e+09/7.8e+09	1.0e+07/1.3e+07	1.9e+05/5.5e+05	9.8e+04/1.6e+05	6.3e+03/7.0e+04	2.0e+07/3.6e+06